

1. *Prove that $1 + 2 + 3 + 4 + \dots + n = \frac{n^2 + n}{2}$*

Base Case : $n = 1$

$$1 = \frac{n^2 + n}{2}$$

$$1 = \frac{1^2 + 1}{2}$$

$$1 = \frac{2}{2}$$

$$1 = 1$$

Proof : $n + 1$

$$1 + 2 + 3 + 4 + \dots + n + (n + 1) = \frac{(n+1)^2 + n+1}{2}$$

$$(1 + 2 + 3 + 4 + \dots + n) + (n + 1) = \frac{n^2 + 2n + 1 + n + 1}{2}$$

$$\frac{n^2 + n}{2} + (n + 1) = \frac{(n+2)(n+1)}{2}$$

$$(n + 1) \left(\frac{n+2}{2} \right) = \frac{(n+2)(n+1)}{2}$$

$$\frac{(n+2)(n+1)}{2} = \frac{(n+2)(n+1)}{2}$$

Thus $S(n + 1)$ is also true.

Since $S(n) \Rightarrow S(n + 1)$ is true, all positive integers n satisfy $S(n) \Rightarrow S(n + 1)$.